

Assignment-3

Q1. Differentiate between **One-Hot Representation** and **Distributed Representation** of words. Why are distributed representations preferred in modern NLP models?

Q2. Explain the **Continuous Bag of Words (CBOW)** and **Skip-Gram** models in Word2Vec. How do they capture semantic relationships between words?

Q3. Discuss the architecture and working of **Convolutional Neural Networks (CNNs)**. Explain the importance of **kernel filters**, **padding**, **stride**, and **max pooling** in feature extraction.

Q4. Compare the architectures and key contributions of **AlexNet**, **GoogLeNet (Inception)**, and **ResNet**. How did these models influence the development of deep convolutional networks?

Q5. What are **Recurrent Neural Networks (RNNs)**? Explain the concept of **Backpropagation Through Time (BPTT)** and how it leads to **exploding** and **vanishing gradient** problems.

Q6. Describe the **Encoder-Decoder model** and its role in **Sequence-to-Sequence learning** tasks such as machine translation. How does **Attention** improve the performance of such models?

Q7. What is the **Attention Mechanism**? Explain different types of attention (e.g., **spatial attention**, **hierarchical attention**) and discuss their applications in **image captioning** and **language modeling**.

Q8.

Define **Generative Adversarial Networks (GANs)**. Explain their **architecture**, **training process**, and mention at least two **real-world applications** of GANs.